HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING

PROJECT REPORT OF

Background Digital Communication

Topic 04: Performance Evaluation of 2x2 MIMO-OFDM System with Alamouti STBC and 16-QAM over Rayleigh Fading Channels

Group 03

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Abstract

The objective of this project is to simulate and evaluate the performance of a 2x2 Multiple-Input Multiple-Output Orthogonal Frequency Division Multiplexing (MIMO-OFDM) communication system. The system utilizes 16-QAM modulation and Alamouti Space-Time Block Coding (STBC) over a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN). Perfect Channel State Information (CSI) is assumed at the receiver to perform Maximum Likelihood detection. The system's performance is quantitatively evaluated using Bit Error Rate (BER) and Symbol Error Rate (SER) as functions of the Signal-to-Noise Ratio (SNR). Simulation results demonstrate the effectiveness of combining OFDM with Alamouti STBC to combat multipath fading and achieve significant diversity gain.

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1 Introduction

The rapid evolution of wireless communication demands robust systems capable of high data rates and reliable transmission over hostile propagation environments. This project simulates a 2x2 MIMO-OFDM communication system using 16-Quadrature Amplitude Modulation (16-QAM) over a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN). Alamouti Space-Time Block Coding (STBC) is employed to provide transmit diversity. The system performance is evaluated by analyzing the Bit Error Rate (BER) and Symbol Error Rate (SER) across various Signal-to-Noise Ratio (SNR) levels.

2 Overview of Technologies Used

This project integrates several fundamental technologies in modern digital communication systems:

- **OFDM (Orthogonal Frequency Division Multiplexing):** Adopted for its high spectral efficiency and robustness against frequency-selective fading. By dividing the bandwidth into orthogonal subcarriers and employing a Cyclic Prefix (CP), OFDM mitigates Inter-Symbol Interference (ISI).
- **16-QAM (Quadrature Amplitude Modulation):** A high-order modulation scheme mapping 4 bits per symbol, significantly improving spectral efficiency compared to lower-order schemes.
- **MIMO and Alamouti STBC:** Multiple-Input Multiple-Output (MIMO) technology using the Alamouti STBC scheme to exploit spatial diversity, combat deep fading, and improve link reliability without requiring extra bandwidth.
- **Rayleigh Fading Channel:** Characterizes environments with rich multipath propagation. Additive White Gaussian Noise (AWGN) is also incorporated to model thermal noise at the receiver.

3 Theoretical Background

3.1 Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal Frequency-Division Multiplexing (OFDM) is a special case of FDM (Frequency Division Multiplexing). In the FDM technique, the total bandwidth of the transmission link is divided into N non-overlapping frequency channels. The signal of each channel is modulated with a subcarrier, and the N channels are multiplexed using frequency division. This results in bandwidth inefficiency; to overcome this drawback of FDM, the OFDM technique was introduced.

This technique divides the frequency band into numerous sub-bands with different carriers, each of which is modulated to transmit a low-rate data stream. The aggregate of these low-rate data streams constitutes the high-rate data stream required for transmission. Furthermore, the carriers employed are orthogonal to each other, allowing their spectra to overlap without causing interference. Consequently, bandwidth utilization becomes significantly more efficient.

3.1.1 Orthogonality Principle

The fundamental concept of OFDM relies on the mathematical orthogonality of its subcarriers. Over a single symbol period T_s, two subcarriers with frequencies f_k and f_m are orthogonal if the integral of their product over the symbol duration equals zero. Mathematically, this continuous-time condition is expressed as:

$$\frac{1}{T_s}\int_0^{T_s} e^{j2\pi f_k t} \times e^{-j2\pi f_m t} dt = \begin{cases} 1, & \text{if } k = m \\ 0, & \text{if } k \neq m \end{cases} \tag{1}$$

To satisfy this condition, the subcarrier spacing Δf is chosen to be the reciprocal of the useful symbol duration $1/T_s$, such that $\Delta f = 1/T_s$. In the frequency domain, the spectrum of each subcarrier follows a sinc function shape. This precise spacing ensures that the peak of one subcarrier aligns with the nulls of all other subcarriers in the frequency domain. Consequently, crosstalk between subcarriers - known as Inter-Carrier Interference (ICI) is theoretically eliminated at the receiver's sampling instants.

3.1.2 Principles of using IFFT and FFT

In practical systems, using thousands of independent oscillators to generate orthogonal subcarriers is hardware-infeasible. Therefore, the Fourier transform is used as an effective alternative.

At the transmitter (using IFFT): Complex data symbols are mapped onto subcarriers in the frequency domain. The inverse fast Fourier transform (IFFT) is performed to convert these symbols to the time domain before transmission. The general formula for the discrete signal $x(n)$ after IFFT is:

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi kn}{N}} \tag{2}$$

where $n = 0, 1, \ldots, N-1$ is the time-domain sample index. The scaling factor $1/\sqrt{N}$ ensures energy normalization.

At the receiver (using FFT): After receiving the time-domain signal, the receiver uses the Fast Fourier Transform (FFT) to convert the signal back to the frequency domain, thereby restoring the original data symbols.

Advantages: IFFT/FFT architecture significantly reduces computational complexity and makes system implementation hardware-feasible at a low cost.

3.1.3 Cyclic Prefix (CP) and ISI Mitigation

Cyclic Prefix (CP) is a guard interval inserted before each symbol to ensure transmission quality in multipath environments. The CP is created by copying the last N_{cp} samples of an OFDM symbol (after the IFFT block) and prepending them to the beginning of that same symbol.

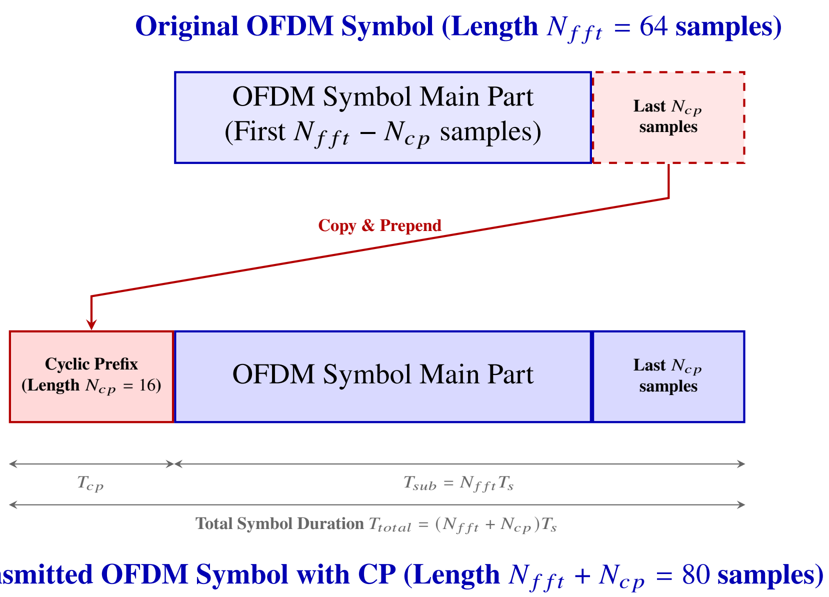

Figure 1: Cyclic Prefix (CP) Insertion Scheme in OFDM Symbol

Specifically, let the time-domain OFDM symbol after IFFT be represented as:

$$\mathbf{x} = [x(0), x(1), \ldots, x(N-1)] \tag{3}$$

The cyclic prefix consists of the last N_{cp} samples: $[x(N - N_{cp}), \ldots, x(N - 1)]$. The transmitted symbol vector with CP becomes:

$$\mathbf{x}_{cp} = [x(N - N_{cp}), \ldots, x(N - 1), x(0), x(1), \ldots, x(N - 1)] \tag{4}$$

The role of the Cyclic Prefix can be detailed as follows:

- **Inter-Symbol Interference (ISI) Mitigation:** In a multipath propagation environment with a maximum channel delay spread of L taps, delayed replicas of the previous OFDM symbol overlap with the current symbol, causing ISI. If the CP length is chosen such that $N_{cp} \geq L - 1$, the delayed components of the previous symbol fall entirely within the CP guard interval and are discarded at the receiver, completely mitigating ISI.

- **Maintaining Orthogonality and Preventing ICI:** Without a CP, the propagation of the signal through a multipath channel corresponds to a linear convolution between the transmitted signal and the channel impulse response. This linear convolution breaks the orthogonality between subcarriers, leading to Inter-Carrier Interference (ICI). By prepending the CP, the linear convolution of the channel over the useful part of the symbol ($n = 0, \ldots, N - 1$) is mathematically transformed into a circular convolution:

$$y(n) = x(n) \circledast h(n) + \eta(n), \quad 0 \leq n \leq N - 1 \tag{5}$$

  where $\circledast$ denotes circular convolution, $h(n)$ is the channel impulse response, and $\eta(n)$ is the noise.

- **Receiver Simplification (One-Tap Equalization):** Due to the Discrete Fourier Transform (DFT) properties, circular convolution in the time domain is equivalent to point-wise multiplication in the frequency domain. Therefore, after taking the FFT at the receiver, the received signal at subcarrier k is given by:

$$Y(k) = H(k)X(k) + N(k) \tag{6}$$

  where $H(k) = \mathrm{FFT}(h)$ is the channel frequency response, and $N(k)$ is the frequency-domain noise. This allows the receiver to perform equalization using a simple division (one-tap equalization) per subcarrier:

$$\hat{X}(k) = \frac{Y(k)}{H(k)} \tag{7}$$

which avoids the need for computationally heavy time-domain equalizers.

3.1.4 Continuous Transmission Model and ISI Analysis

In a continuous transmission system, a sequence of P OFDM symbols is transmitted sequentially in the time domain. Let $x_p(n)$ represent the p-th time-domain OFDM symbol with CP of length N_{cp}, where $p \in \{1, 2, \ldots, P\}$ and $n \in \{0, 1, \ldots, N_{fft} + N_{cp} - 1\}$. The concatenated continuous transmit signal for a given antenna can be represented as:

$$s(t) = \sum_{p=1}^{P} x_p(t - (p-1)(N_{fft} + N_{cp})) \tag{8}$$

When this continuous stream propagates through a multipath Rayleigh fading channel with impulse response $h(t) = \sum_{l=0}^{L_{ch}-1} h_l \delta(t-l)$, the received signal $r(t)$ is the continuous convolution:

$$r(t) = (s * h)(t) + \eta(t) = \sum_{l=0}^{L_{ch}-1} h_l s(t-l) + \eta(t) \tag{9}$$

Let us focus on the p-th received OFDM symbol interval, corresponding to $t = (p - 1)(N_{fft} + N_{cp}) + n$ for $n \in \{0, 1, \ldots, N_{fft} + N_{cp} - 1\}$. The received samples $r_p(n)$ can be written as:

$$r_p(n) = \sum_{l=0}^{L_{ch}-1} h_l s((p-1)(N_{fft} + N_{cp}) + n - l) + \eta_p(n) \tag{10}$$

For $n < L_{ch} - 1$, the term $n - l$ can be negative, meaning that the received samples depend on the previous symbol $s((p-1)(N_{fft} + N_{cp}) + n - l) = x_{p-1}(N_{fft} + N_{cp} + n - l)$. This represents the Inter-Symbol Interference (ISI) from symbol $p - 1$ leaking into the current symbol p.

At the receiver, the first N_{cp} samples of the received symbol block $r_p(n)$ are discarded.

- **Sufficient CP ($N_{cp} \geq L_{ch} - 1$):** Since the ISI only affects the first $L_{ch} - 1$ samples of the block, discarding the first N_{cp} samples completely eliminates the ISI. The remaining N_{fft} samples $y_p(m) = r_p(m + N_{cp})$ for $m \in \{0, 1, \ldots, N_{fft} - 1\}$ contain no contributions from $x_{p-1}(n)$ and represent a clean circular convolution of the current symbol $x_p(n)$ with the channel $h(n)$.

- **Insufficient CP ($N_{cp} < L_{ch} - 1$):** If the CP is shorter than the channel delay spread, the ISI leaks into the useful FFT window. The received samples in the FFT window

$y_p(m) = r_p(m + N_{cp})$ for $m < L_{ch} - 1 - N_{cp}$ will be corrupted by the tail of the previous symbol $x_{p-1}(n)$, resulting in a permanent error floor at high SNR.

3.2 Quadrature Amplitude Modulation (16-QAM)

3.2.1 Theory of 16-QAM Modulation

Quadrature Amplitude Modulation (QAM) is a digital modulation technique that transmits information by varying both the amplitude and phase of a carrier signal. The transmitted signal consists of two orthogonal components: In-phase component (I) and Quadrature component (Q).

For a 16-QAM system, the modulation order is $M = 16$. The number of bits carried by each symbol is $k = \log_2(M) = 4$ bits/symbol. Therefore, each transmitted symbol carries 4 bits of information.

The passband signal is given by:

$$s(t) = I\cos(2\pi f_c t) - Q\sin(2\pi f_c t) \tag{11}$$

where I is the In-phase component, Q is the Quadrature component, and f_c is the Carrier frequency.

The amplitude levels used in both I and Q branches are $I, Q \in \{-3, -1, +1, +3\}$. Since each axis contains four amplitude levels, the total number of constellation points is $N = 4 \times 4 = 16$. Thus, a 16-point constellation is obtained.

3.2.2 Average Symbol Energy Normalization

The average energy of the in-phase component is calculated as:

$$\mathbb{E}[I^2] = \frac{(-3)^2 + (-1)^2 + (+1)^2 + (+3)^2}{4} = \frac{9 + 1 + 1 + 9}{4} = 5 \tag{12}$$

Similarly, $\mathbb{E}[Q^2] = 5$. Therefore, the average symbol energy becomes:

$$E_s = \mathbb{E}[I^2] + \mathbb{E}[Q^2] = 5 + 5 = 10 \tag{13}$$

To normalize the constellation such that $E_s = 1$, all constellation points are multiplied by $1/\sqrt{10}$. Hence, the normalized transmitted symbol is:

$$s = \frac{I + jQ}{\sqrt{10}} \tag{14}$$

This normalization is used in the MATLAB implementation and simplifies BER and SER calculations.

3.2.3 16-QAM Constellation Diagram and Gray Bit Mapping

Gray coding is employed so that adjacent constellation points differ by only one bit. The first two bits determine the I component, while the last two bits determine the Q component.

Table 1: Constellation Coordinates and Gray Mapping

Q \I	**I = -3 (00)**	**I = -1 (01)**	**I = +1 (11)**	**I = +3 (10)**
Q = +3 (10)	(-3, 3) / 0010	(-1, 3) / 0110	(1, 3) / 1110	(3, 3) / 1010
Q = +1 (11)	(-3, 1) / 0011	(-1, 1) / 0111	(1, 1) / 1111	(3, 1) / 1011
Q = -1 (01)	(-3,-1) / 0001	(-1,-1) / 0101	(1,-1) / 1101	(3,-1) / 1001
Q = -3 (00)	(-3,-3) / 0000	(-1,-3) / 0100	(1,-3) / 1100	(3,-3) / 1000

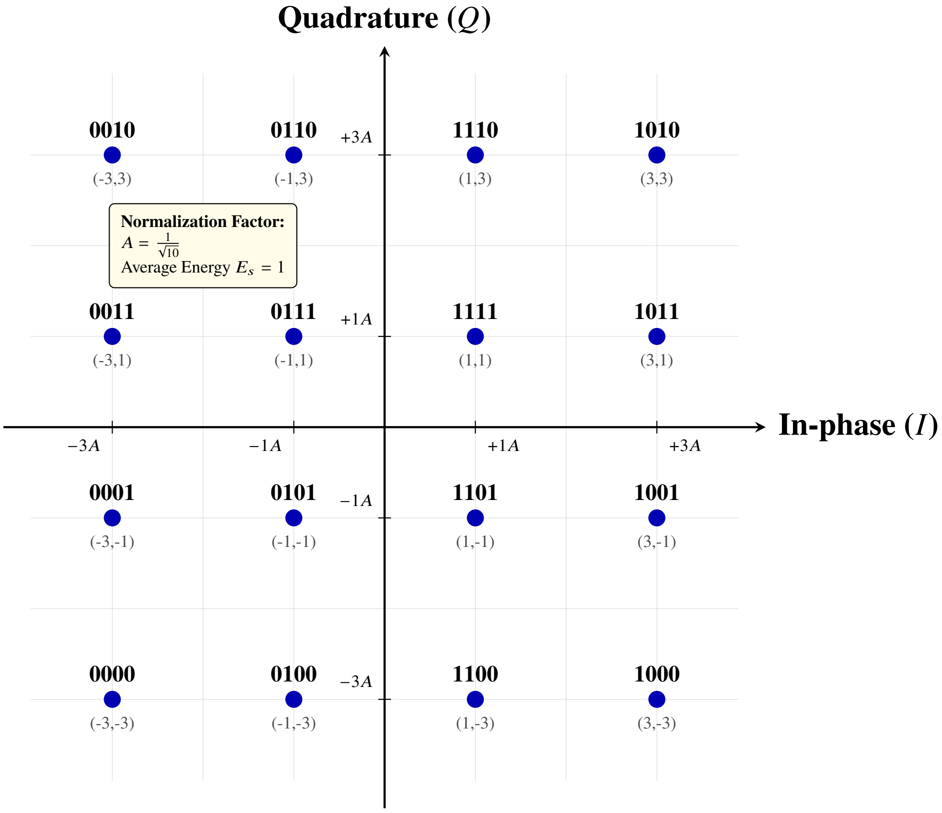

Figure 2: 16-QAM Constellation Diagram with Gray Mapping

To clarify the mapping rule, several examples are presented:

- **Example 1 (Input bits** 0000**):** The first two bits 00 correspond to $I = -3$, and the last two bits 00 correspond to $Q = -3$. The resulting complex symbol is $s = \frac{-3-j3}{\sqrt{10}}$.
- **Example 2 (Input bits** 1111**):** The first two bits 11 correspond to $I = +1$, and the last two bits 11 correspond to $Q = +1$. The resulting complex symbol is $s = \frac{1+j1}{\sqrt{10}}$.
- **Example 3 (Input bits** 1010**):** The first two bits 10 correspond to $I = +3$, and the last two bits 10 correspond to $Q = +3$. The resulting complex symbol is $s = \frac{3+j3}{\sqrt{10}}$.

3.2.4 Euclidean Distance Demodulation

After transmission through the Rayleigh fading channel and AWGN, the received symbol is represented as $r = I_r + jQ_r$. Let the m-th constellation point be $S_m = I_m + jQ_m$. The Euclidean distance between the received symbol and the constellation point is:

$$d_m = \sqrt{(I_r - I_m)^2 + (Q_r - Q_m)^2} \tag{15}$$

To reduce computational complexity, the receiver usually employs the squared Euclidean distance. The Maximum Likelihood (ML) decision rule is:

$$\hat{S} = \arg\min_{S_m} \left((I_r - I_m)^2 + (Q_r - Q_m)^2 \right) \tag{16}$$

The constellation point with the minimum Euclidean distance is selected as the detected symbol and then converted back into the corresponding 4-bit sequence.

3.3 MIMO (Multiple Input Multiple Output) and Alamouti STBC

MIMO is a key technique in modern wireless communication systems that utilizes multiple antennas at both the transmitter and receiver. By transmitting and receiving multiple signals simultaneously over the same frequency band, MIMO systems exploit the spatial dimension to achieve:

- **Spatial Multiplexing Gain:** Transmitting independent data streams from different antennas to linearly scale the system capacity with the number of antennas.
- **Diversity Gain:** Transmitting the same or related signals through multiple independent fading paths to improve transmission reliability and combat channel fading.

A 2×2 MIMO system consists of $N_T = 2$ transmit antennas and $N_R = 2$ receive antennas, creating four spatial communication links. In a flat-fading or subcarrier-level frequency-domain representation, the received signal vector $\mathbf{y} = [y_1, y_2]^T$ is modeled as:

$$\mathbf{y} = \mathbf{Hs} + \boldsymbol{\eta} \tag{17}$$

where $\mathbf{s} = [s_1, s_2]^T$ is the transmitted symbol vector, $\boldsymbol{\eta} = [\eta_1, \eta_2]^T$ is the additive white Gaussian noise (AWGN) vector, and $\mathbf{H}$ is the 2×2 channel matrix:

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \tag{18}$$

Here, h_{ij} represents the complex channel path gain from transmit antenna j to receive antenna i.

3.3.1 MIMO-OFDM Integration

When MIMO is combined with OFDM, the wideband frequency-selective MIMO channel is divided into N_{fft} parallel, orthogonal narrowband flat-fading MIMO subchannels. For each subcarrier $k \in \{0, 1, \ldots, N_{fft} - 1\}$, the relationship between the transmitted symbol vector $\mathbf{s}(k) = [s_1(k), s_2(k)]^T$ and the received vector $\mathbf{y}(k) = [y_1(k), y_2(k)]^T$ is given by:

$$\mathbf{y}(k) = \mathbf{H}(k)\mathbf{s}(k) + \boldsymbol{\eta}(k) \tag{19}$$

where $\mathbf{H}(k)$ is the estimated channel response matrix at subcarrier k, containing the channel gains $h_{ij}(k)$ between transmit antenna j and receive antenna i at frequency subcarrier k.

3.3.2 Practical Transmitter Components: Channel Coding and Interleaving

In practical MIMO-OFDM transmitters (such as in IEEE 802.11 or LTE), the input binary stream is processed through several additional stages before being mapped into symbols:

- **Channel Encoder:** Adds redundant bits (using convolutional codes, Turbo codes, or LDPC codes) to enable forward error correction (FEC) at the receiver.
- **Interleaver (IL):** Permutes the coded bits in a pseudo-random order. This is crucial in OFDM systems since deep fading on certain subcarriers can corrupt consecutive bits. Interleaving spreads these errors across the entire block, transforming burst errors into random, single-bit errors that the channel decoder can easily correct.

While these blocks are essential for commercial systems to achieve the required Quality of Service (QoS), they are omitted in this simulation to focus strictly on evaluating the raw physical layer performance of the 16-QAM MIMO-OFDM scheme with Alamouti STBC under raw fading channels.

3.3.3 Alamouti STBC Encoding

To exploit diversity gain without needing multiple RF chains or channel estimation at the transmitter, the Alamouti Space-Time Block Coding (STBC) scheme is employed. The code operates across two consecutive OFDM symbols (or time slots) $t = 1, 2$.

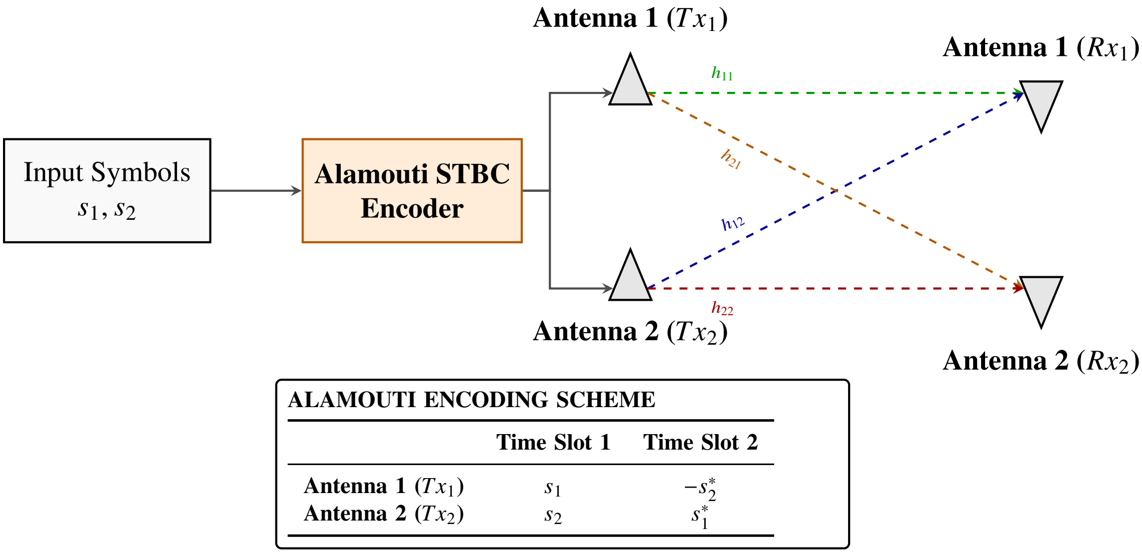

Figure 3: Alamouti Space-Time Block Coding Transmission Scheme

The implemented system uses a 2x2 Alamouti Space-Time Block Code (STBC). For two consecutive symbols s_1 and s_2, the transmission matrix over two time slots is:

$$\begin{bmatrix} s_1 & -s_2^* \\ s_2 & s_1^* \end{bmatrix} \tag{20}$$

where $(*)$ denotes complex conjugation. This coding scheme provides transmit diversity and improves performance in fading environments.

3.3.4 Signal Model at Receiver

In the first time slot, Antenna 1 transmits $s_1(k)$ and Antenna 2 transmits $s_2(k)$ on subcarrier k. The received signals at receive antenna $i \in \{1, 2\}$ in time slot 1 are:

$$y_{i1}(k) = h_{i1}(k)s_1(k) + h_{i2}(k)s_2(k) + \eta_{i1}(k) \tag{21}$$

which can be written in matrix form as:

$$\begin{bmatrix} y_{11}(k) \\ y_{21}(k) \end{bmatrix} = \begin{bmatrix} h_{11}(k) & h_{12}(k) \\ h_{21}(k) & h_{22}(k) \end{bmatrix} \begin{bmatrix} s_1(k) \\ s_2(k) \end{bmatrix} + \begin{bmatrix} \eta_{11}(k) \\ \eta_{21}(k) \end{bmatrix} \tag{22}$$

In the second time slot, Antenna 1 transmits $-s_2^*(k)$ and Antenna 2 transmits $s_1^*(k)$ on subcarrier k. The received signals at receive antenna $i \in \{1, 2\}$ in time slot 2 are:

$$y_{i2}(k) = -h_{i1}(k)s_2^*(k) + h_{i2}(k)s_1^*(k) + \eta_{i2}(k) \tag{23}$$

which in matrix form is:

$$\begin{bmatrix} y_{12}(k) \\ y_{22}(k) \end{bmatrix} = \begin{bmatrix} h_{11}(k) & h_{12}(k) \\ h_{21}(k) & h_{22}(k) \end{bmatrix} \begin{bmatrix} -s_2^*(k) \\ s_1^*(k) \end{bmatrix} + \begin{bmatrix} \eta_{12}(k) \\ \eta_{22}(k) \end{bmatrix} \tag{24}$$

3.3.5 Alamouti STBC Decoding

To normalize the total transmit power such that the sum of the power of the two transmit antennas is equal to 1, a power scale factor $t = \text{txScale} = 1/\sqrt{N_t} = 1/\sqrt{2}$ is applied at the transmitter. At the receiver, assuming Perfect Channel State Information (CSI), the channel estimation is represented using the effective channel response $g_{ij}(k) = \text{txScale} \cdot h_{ij}(k) = \frac{1}{\sqrt{2}} h_{ij}(k)$.

The transmitted symbols $s_1(k)$ and $s_2(k)$ are estimated by combining the received signals across both receive antennas and both time slots using these effective channel gains. The combination equations are:

$$\tilde{s}_1(k) = \sum_{i=1}^{2} \left(g_{i1}^*(k) y_{i1}(k) + g_{i2}(k) y_{i2}^*(k) \right) \tag{25}$$

$$\tilde{s}_2(k) = \sum_{i=1}^{2} \left(g_{i2}^*(k) y_{i1}(k) - g_{i1}(k) y_{i2}^*(k) \right) \tag{26}$$

To compute the final decision variables for Maximum Likelihood (ML) detection, these combined signals are normalized by the total power of the active effective channels:

$$\hat{s}_1(k) = \frac{\tilde{s}_1(k)}{\sum_{i=1}^{2} \left(|g_{i1}(k)|^2 + |g_{i2}(k)|^2 \right)} \tag{27}$$

$$\hat{s}_2(k) = \frac{\tilde{s}_2(k)}{\sum_{i=1}^{2} \left(|g_{i1}(k)|^2 + |g_{i2}(k)|^2 \right)} \tag{28}$$

These decision variables $\hat{s}_1(k)$ and $\hat{s}_2(k)$ are then passed to the Euclidean distance detector. Because both the received signals $y_{ij}(k)$ and the effective channel gains $g_{ij}(k)$ contain the factor txScale, the term txScale^2 appears in both the numerator and the denominator and cancels out perfectly. Therefore, the recovered symbols are restored to their original constellation scale without any amplitude shrinkage.

4 System Block Diagram and Operation

4.1 System Block Diagram

To understand the interaction between the components, the complete block diagram of the simulated 2×2 MIMO-OFDM communication system is shown below.

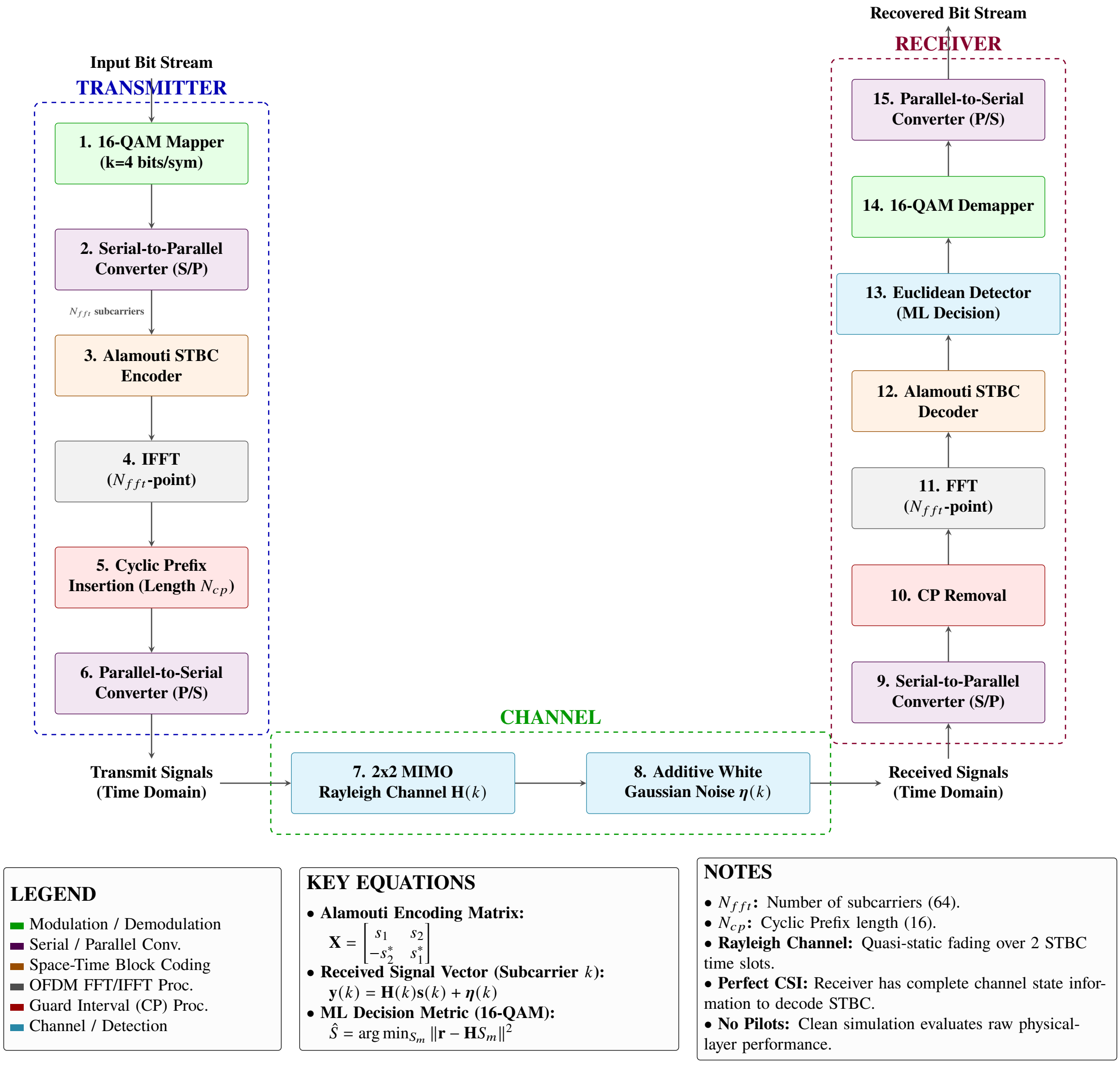

Figure 4: Detailed System Block Diagram of the 2×2 MIMO-OFDM System

The processing chain is structured into three main segments:

- **Transmitter:**
  1. *Bit Generator:* Generates a stream of pseudo-random binary bits.
  2. *16-QAM Mapper:* Groups the bits into 4-bit symbols and maps them onto the complex 16-QAM constellation (serial symbol stream).
  3. *Serial to Parallel (S/P) Converter:* Splits the serial symbol stream into parallel blocks of length N_{fft} to map them onto the orthogonal subcarriers for each transmit antenna path (s_1 and s_2).
  4. *Alamouti STBC Encoder:* Encodes the parallel symbols in pairs (s_1, s_2) across two spatial paths (Antenna 1 and Antenna 2) and two time slots.
  5. *OFDM Modulator (IFFT + CP):* Applies an N_{fft}-point Inverse Fast Fourier Transform (IFFT) to convert frequency subcarriers to time-domain samples, and prepends a Cyclic Prefix (CP) of length N_{cp} to mitigate multipath interference.
  6. *Parallel to Serial (P/S) Converter:* Converts the parallel time-domain OFDM symbols back to a serial sample sequence for transmission over the air via the antennas.

- **Transmission Channel:**
  1. *Multipath Rayleigh Fading:* Simulates a multipath channel environment using a 6-tap exponentially decaying power delay profile. The fading is assumed to be quasi-static (constant over two Alamouti symbol periods).
  2. *Additive White Gaussian Noise (AWGN):* Models thermal noise at each receive antenna.

- **Receiver:**
  1. *Serial to Parallel (S/P) Converter:* Buffers the received serial time-domain signal samples into parallel blocks of length $N_{fft} + N_{cp} = 80$ to prepare for demodulation.
  2. *OFDM Demodulator (Remove CP + FFT):* Discards the CP and applies an N_{fft}-point Fast Fourier Transform (FFT) to return the signals to the frequency domain (parallel subcarriers).
  3. *Alamouti STBC Decoder:* Combines received signals across the two time slots and two antennas using channel estimates (Perfect CSI) to separate and estimate the transmitted symbols.

4. *Euclidean Detector:* Computes the squared Euclidean distance between the estimated symbols and the normalized 16-QAM constellation points.
5. *16-QAM Demapper:* Recovers the 4-bit binary sequences from the detected symbols.
6. *Parallel to Serial (P/S) Converter:* Restores the single serial bit stream from the parallel decoded bits for final Bit Error Rate (BER) evaluation.

4.2 Role of Serial-to-Parallel (S/P) and Parallel-to-Serial (P/S) Converters

In the physical communication system, data is generated as a serial bit stream, and the channel propagates signals as a continuous serial wave in time. However, OFDM is inherently a multi-carrier system where symbols are mapped onto parallel subcarriers, and STBC processes symbols in parallel blocks. Thus, S/P and P/S converters are essential interface blocks between serial transmission media and parallel processing modules:

- **Transmitter S/P (Step 2 in diagram):** Converts the serial stream of 16-QAM symbols into parallel blocks. This is required because the IFFT block processes all $N_{fft} = 64$ subcarriers simultaneously.

- **Transmitter P/S (Step 6 in diagram):** Converts the parallel time-domain samples of the OFDM symbol (length 80) into a serial sequence of samples, which are transmitted one by one over the antenna.

- **Receiver S/P (Step 9 in diagram):** Receives the serial time-domain samples from the antenna and groups them into parallel blocks of size 80 so that the CP can be removed and the 64-sample block can be fed into the FFT.

- **Receiver P/S (Step 15 in diagram):** Converts the parallel decoded bits back into a single serial bit stream to compare with the transmitter's input stream for BER calculation.

In the MATLAB simulation environment (`testthongtinso.m`), these hardware blocks do not exist as separate components but are implemented implicitly via array slicing and matrix reshaping:

- **Transmitter S/P:** Implemented by partitioning the serial symbol vector `txSymbols` into parallel components `s1 = txSymbols(1:Nfft).'` and `s2 = txSymbols(Nfft+1:end`

- **Transmitter P/S:** The output of `ofdm_mod` is a 1x80 vector, which represents the serial sample sequence. The channel convolution `conv(x, h)` processes this vector sequentially, simulating serial transmission.

- **Receiver S/P:** Implemented by extracting a slice of the received vector `rxTime(Ncp+1:Ncp+Nf` in `ofdm_demod`, which converts a portion of the serial received stream into a block of 64 samples.

- **Receiver P/S:** Done by concatenating parallel estimated symbols into a column vector `rxSymbols = [s1_hat.'; s2_hat.']` and reshaping the bit matrix back to a 1D column vector `bits = reshape(bitGroups.', [], 1)` in `qam16_demod`.

Thus, S/P and P/S conversions are fully represented in the simulation logic through standard index-slicing and data-reshaping operations, which correspond exactly to their physical hardware counterparts.

4.3 System Operation

1. **Bit Generation:** A random binary bit stream is generated.

2. **16-QAM Mapping:** Every four bits are mapped into one normalized 16-QAM symbol according to the Gray coding rule.

3. **Serial to Parallel (S/P) Conversion:** Splits the serial QAM symbols into parallel subcarriers for the transmit antennas.

4. **Alamouti STBC Encoding:** The symbols are encoded using Alamouti STBC to exploit spatial diversity.

5. **OFDM Modulation:** The encoded symbols are transformed into the time domain using an Inverse Fast Fourier Transform (IFFT). A Cyclic Prefix (CP) is added to mitigate inter-symbol interference.

6. **Parallel to Serial (P/S) Conversion:** Converts the parallel time-domain samples of the OFDM symbol into a serial sequence for transmission.

7. **Channel Transmission:** The OFDM signal propagates through a multipath Rayleigh fading channel and is corrupted by Additive White Gaussian Noise (AWGN).

8. **Serial to Parallel (S/P) Conversion:** The received serial time-domain signal is buffered into parallel blocks of samples at the receiver.

9. **OFDM Demodulation:** At the receiver, the cyclic prefix is removed and a Fast Fourier Transform (FFT) is performed to return to the frequency domain (parallel subcarriers).

10. **Alamouti STBC Decoding:** The receiver combines signals from multiple antennas and estimates the transmitted symbols using Perfect CSI.

11. **Euclidean Distance Detection:** The Euclidean distance between the received symbol and all possible constellation points is calculated. The constellation point with the minimum distance is selected.

12. **16-QAM Demapping:** The detected symbol is converted back into the corresponding binary sequence using the Gray mapping rule.

13. **Parallel to Serial (P/S) Conversion:** Restores the single serial bit stream from the parallel decoded bits.

14. **Performance Evaluation:** The recovered bit stream is compared with the transmitted bit stream to evaluate BER and SER performance.

5 Experimental Setup

The simulation framework was developed using a MATLAB-based environment to evaluate the performance of a 2x2 MIMO-OFDM system. The proposed system models the transmission of a random bit stream through a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN).

5.1 Simulation Parameters

To ensure reproducibility and clarity, the main parameters used in the simulation are summarized in Table 2.

Table 2: Simulation Parameters

Parameter	Value
Number of Transmit Antennas	2
Number of Receive Antennas	2
MIMO Scheme	Alamouti STBC
FFT Size (N)	64
Cyclic Prefix (CP) Length	16
Modulation Scheme	16-QAM
Bits per Symbol	4
Channel Type	Multipath Rayleigh Fading
Number of Channel Taps	6
Noise Model	AWGN
E_b/N_0 Range	0 dB to 30 dB
Performance Metrics	BER and SER
Channel State Information	Perfect CSI assumed at receiver

5.2 Evaluation Scenario

The simulation is repeated for E_b/N_0 values from 0 dB to 30 dB. For each value, a large number of random bits are transmitted through the 2x2 MIMO-OFDM system. The corresponding BER and SER values are collected and plotted on a logarithmic scale. These curves are then used to evaluate how the system performance changes as the signal-to-noise ratio increases. The expected result is that both BER and SER decrease as E_b/N_0 increases. The use of Alamouti STBC is expected to improve system reliability by providing spatial diversity, while OFDM helps reduce the effect of multipath propagation by converting the frequency-selective channel into multiple narrowband subchannels.

6 Results and Discussion

The performance of the 2x2 MIMO-OFDM system utilizing Alamouti Space-Time Block Coding (STBC) and 16-QAM modulation over a Rayleigh fading channel with additive white Gaussian noise (AWGN) is evaluated by measuring the Bit Error Rate (BER) and Symbol Error Rate (SER) across various Signal-to-Noise Ratio (SNR) levels.

6.1 Simulation Results

The obtained simulation results illustrate the variations of BER and SER as functions of E_b/N_0. The corresponding performance curves are plotted in Figure 5. To provide quantitative precision, specific performance data points extracted from the simulation with a step size of 2 dB are presented in Table 3.

Table 3: BER and SER Performance Data of the 16-QAM MIMO-OFDM System

E_b/N_0 (dB)	Bit Error Rate (BER)	Symbol Error Rate (SER)
0	9.11283e-02	3.19521e-01
2	5.59388e-02	2.03760e-01
4	2.97852e-02	1.11914e-01
6	1.36120e-02	5.18828e-02
8	4.64779e-03	1.80755e-02
10	1.39518e-03	5.45573e-03
12	4.33594e-04	1.69792e-03
14	7.42187e-05	2.96875e-04
16	1.75781e-05	6.77083e-05
18	6.51042e-07	2.60417e-06
20	6.51042e-07	2.60417e-06
22	6.51042e-07	2.60417e-06
24	0.00000e+00	0.00000e+00
26	0.00000e+00	0.00000e+00
28	0.00000e+00	0.00000e+00
30	0.00000e+00	0.00000e+00

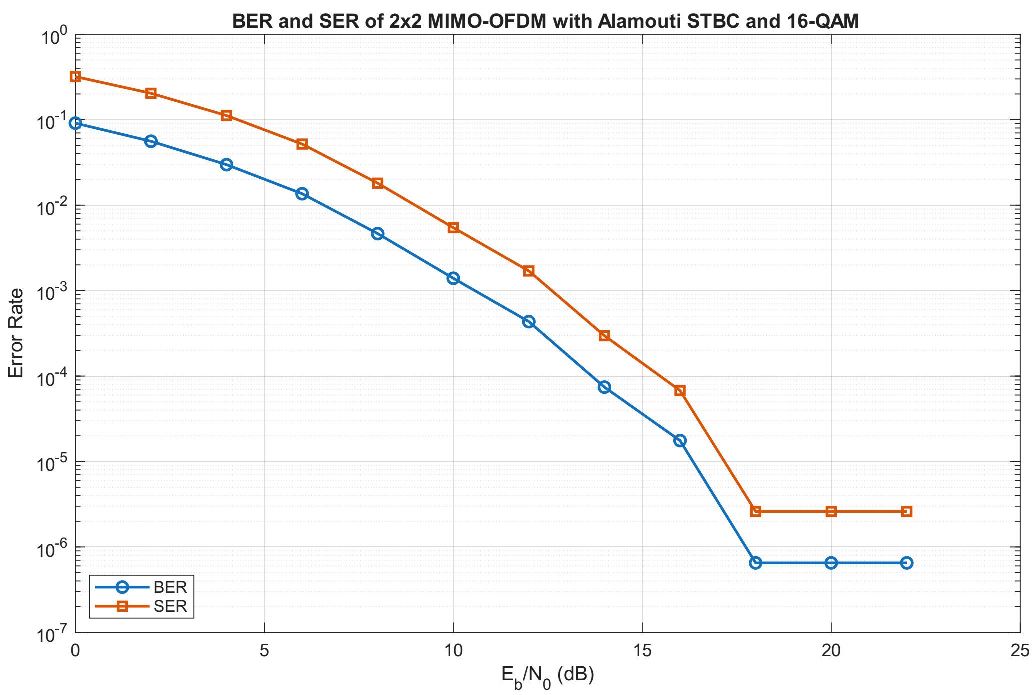

Figure 5: BER and SER Performance of the 2x2 MIMO-OFDM System with Alamouti STBC and 16-QAM

To analyze the performance of the system under physical multi-symbol conditions, a continuous transmission model is simulated. The bit error rate (BER) curves obtained for three different cyclic prefix lengths ($N_{cp} \in \{16, 4, 0\}$) under a 6-tap Rayleigh fading channel are presented in Table 4 and plotted in Figure 6.

Table 4: BER Performance with Different CP Lengths under Continuous Transmission

E_b/N_0 **(dB)**	**Sufficient CP** ($N_{cp} = 16$)	**Insufficient CP** ($N_{cp} = 4$)	**No CP** ($N_{cp} = 0$)
0	9.31083e-02	9.19773e-02	1.00272e-01
4	2.94128e-02	3.06538e-02	3.95167e-02
8	5.18065e-03	4.93648e-03	1.20423e-02
12	4.01979e-04	4.08333e-04	3.45750e-03
16	1.81250e-05	2.05729e-05	1.67828e-03
20	2.86458e-07	1.69271e-06	1.06664e-03
24	0.00000e+00	0.00000e+00	9.14505e-04
28	0.00000e+00	0.00000e+00	8.57891e-04
30	0.00000e+00	0.00000e+00	9.05703e-04

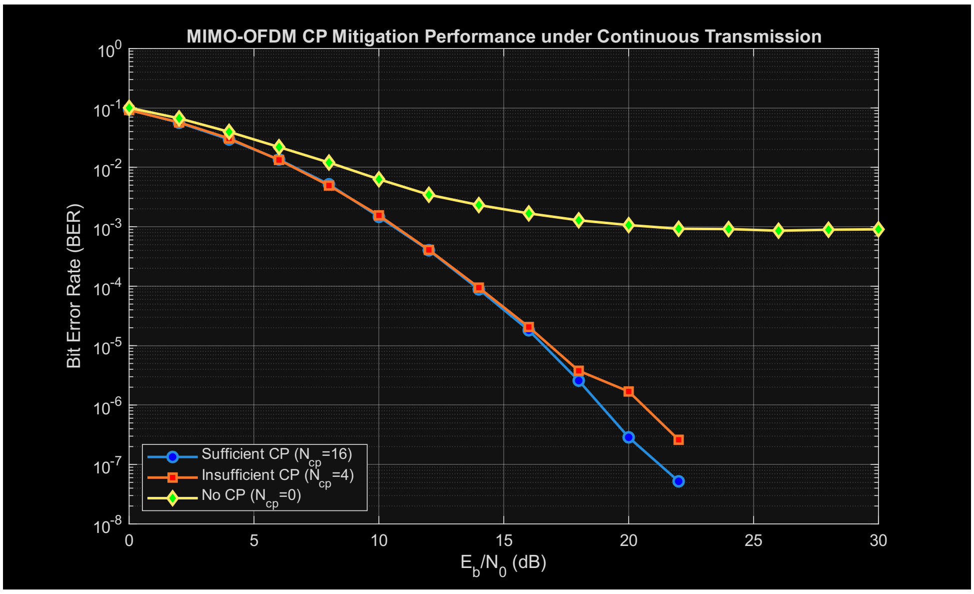

Figure 6: BER Performance Comparison of 2x2 MIMO-OFDM with Different CP Lengths under Continuous Transmission

6.2 Discussion

6.2.1 General Performance Trend

The simulation results show that both BER and SER decrease as E_b/N_0 increases. At low E_b/N_0, the received signal is strongly affected by Rayleigh fading and AWGN, resulting in relatively high error rates. As E_b/N_0 increases, the noise effect becomes weaker, so the detection accuracy improves and the BER/SER values decrease significantly.

6.2.2 BER and SER Comparison

The SER values are higher than the BER values because each 16-QAM symbol carries 4 bits. A symbol error occurs when at least one bit in the symbol is detected incorrectly. However, the use of Gray coding helps reduce the number of bit errors when the received symbol is mistaken for a neighboring constellation point.

6.2.3 Diversity Gain from Alamouti STBC

The use of Alamouti STBC in the 2x2 MIMO system provides spatial diversity. By transmitting coded symbols through two transmit antennas and combining the received signals from two receive antennas, the system reduces the impact of deep fading. This explains the rapid decrease of BER and SER as E_b/N_0 increases.

6.2.4 Effect of CP Length under Continuous Transmission

The continuous transmission simulation results clearly demonstrate the vital role of the Cyclic Prefix (CP) in mitigating multipath fading and Inter-Symbol Interference (ISI):

- **Sufficient CP ($N_{cp} = 16 \geq L_{ch} - 1$):** When the CP is longer than the maximum channel delay spread (which is $L_{ch} - 1 = 5$ samples), the ISI from previous symbols falls completely inside the CP guard interval. Since the CP is discarded at the receiver, the remaining FFT window is free from any ISI. Furthermore, the linear convolution of the channel over the symbol is mathematically transformed into a circular convolution, preserving the subcarrier orthogonality and eliminating ICI. Consequently, the BER drops cleanly down to zero at high SNR without experiencing an error floor.

- **Insufficient CP ($N_{cp} = 4 < L_{ch} - 1$):** When the CP is shorter than the channel delay spread, a portion of the channel's multipath components (specifically the 5th tap) spills past the CP and corrupts the start of the FFT window. This causes minor ISI and ICI. At high SNR (e.g., 20 dB), the BER is 1.69×10^{-6}, which is approximately 6 times higher than that of the sufficient CP case (2.86×10^{-7}). However, because the channel profile decays exponentially ($pdp_l = e^{-l}$), the energy of the late taps is extremely small ($e^{-5} \approx 0.0067$), so the resulting ISI/ICI is weak. Thus, the error rate is still able to descend further at very high SNR without forming a prominent floor in the simulated range.

- **No CP ($N_{cp} = 0$):** Without any guard interval, the receiver suffers from severe block-to-block ISI and subcarrier orthogonality loss (ICI) across all channel taps.

This self-interference behaves like additive noise that scales proportionally with the transmit signal power. As a result, the BER flattens out starting at 16 dB and bottlenecks at a permanent **error floor of approximately** 9.0×10^{-4}. This error floor cannot be resolved by increasing the transmit power (SNR), proving that the CP is essential for the basic operation of OFDM in multipath environments.

6.2.5 Simulation Resolution Limit in the High-E_b/N_0 Region

In the high-E_b/N_0 region (specifically from 18 dB to 22 dB), a flattening of the BER and SER curves can be observed at exactly 6.51×10^{-7} and 2.60×10^{-6}, respectively. As the simulation assumes perfect channel state information (CSI) at the receiver, this is not a physical error floor. Rather, it represents the **resolution limit of the Monte Carlo simulation** due to the finite number of transmitted bits.

With the configured $N_{blocks} = 3000$, the total number of transmitted bits per SNR point is $1,536,000$. At these high SNR levels, exactly **one bit error** (and correspondingly one symbol error) occurs across the entire simulation, causing the calculated error rates to bottleneck at a constant minimum measurable value ($1/1,536,000 \approx 6.51 \times 10^{-7}$). Once E_b/N_0 reaches 24 dB, the system becomes entirely error-free (0 errors), causing the BER and SER to drop to strictly zero. To accurately capture and observe a smoother waterfall curve for error probabilities below this threshold, the number of simulation blocks must be significantly increased (e.g., $N_{blocks} \geq 10^5$).

6.2.6 Overall Evaluation

Overall, the results confirm that the 2x2 MIMO-OFDM system using Alamouti STBC and 16-QAM achieves good performance over a multipath Rayleigh fading channel. The performance improvement mainly comes from the diversity gain of STBC and the ability of OFDM to handle multipath propagation.

7 Conclusion

This project presented the simulation and performance evaluation of a 2x2 MIMO-OFDM communication system using 16-QAM modulation and Alamouti Space-Time Block Coding (STBC) over a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN). Perfect Channel State Information (CSI) was assumed at the receiver to perform Maximum Likelihood detection.

The simulation results demonstrated that both Bit Error Rate (BER) and Symbol Error Rate (SER) decrease as the Signal-to-Noise Ratio (SNR) increases. The implemented architecture improves spectral efficiency, provides spatial diversity gain, and enhances communication reliability in wireless fading environments. The use of OFDM effectively handles multipath propagation by eliminating inter-symbol interference through the use of a cyclic prefix. Overall, the project provided practical insights into the behavior of modern wireless communication systems.

8 References

1. J. G. Proakis, *Digital Communications*, 5th ed., McGraw-Hill, 2008.
2. S. Haykin, *Digital Communication Systems*, Wiley, 2013.
3. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed., Prentice Hall, 2002.

A MATLAB Simulation Source Code

The following complete and self-contained MATLAB script was used to perform the Monte Carlo simulation of the 2x2 MIMO-OFDM system with Alamouti STBC and 16-QAM modulation over Rayleigh fading channels:

```matlab
clc;
clear;
close all;

%% ============================================================
%  2x2 MIMO-OFDM with Alamouti STBC over multipath Rayleigh channel
%  Modulation: 16-QAM
%  Metrics: BER and SER versus Eb/N0
%  No Communications Toolbox required
% ============================================================

rng(1);                         % For reproducible results

%% ---------------- Simulation parameters ----------------
Nt = 2;                          % Number of transmit antennas
Nr = 2;                          % Number of receive antennas

M = 16;                          % 16-QAM
k = log2(M);                     % Bits per QAM symbol = 4

Nfft = 64;                       % Number of OFDM subcarriers
Ncp = 16;                        % Cyclic prefix length
Lch = 6;                         % Number of Rayleigh multipath taps

EbN0_dB = 0:2:30;                % Eb/N0 range in dB
Nblocks = 3000;                  % Number of Alamouti-OFDM blocks per Eb/N0
txScale = 1 / sqrt(Nt);          % Normalize total transmit power

BER = zeros(size(EbN0_dB));
SER = zeros(size(EbN0_dB));

% Exponential power delay profile (PDP)
pdp = exp(-(0:Lch-1));
pdp = pdp / sum(pdp);            % Normalize total channel power to 1

%% ============================================================
```

```
%                       Main Eb/N0 simulation loop
% ============================================================

for isnr = 1:length(EbN0_dB)
    EbN0_linear = 10^(EbN0_dB(isnr)/10);
    noiseVar = 1 / (k * EbN0_linear); % Noise variance per complex sample

    totalBitErrors = 0;
    totalSymErrors = 0;
    totalBits = 0;
    totalSymbols = 0;

    for blk = 1:Nblocks
        % 1. Generate random bits
        NsymPerBlock = 2 * Nfft;
        NbitsPerBlock = NsymPerBlock * k;
        txBits = randi([0 1], NbitsPerBlock, 1);

        % 2. 16-QAM mapping
        txSymbols = qam16_mod(txBits);
        s1 = txSymbols(1:Nfft).';
        s2 = txSymbols(Nfft+1:end).';

        % 3. Alamouti STBC encoding in frequency domain
        X_tx1_slot1 = txScale * s1;
        X_tx2_slot1 = txScale * s2;
        X_tx1_slot2 = txScale * (-conj(s2));
        X_tx2_slot2 = txScale * ( conj(s1));

        % 4. OFDM modulation: IFFT + cyclic prefix
        x_tx1_slot1 = ofdm_mod(X_tx1_slot1, Nfft, Ncp);
        x_tx2_slot1 = ofdm_mod(X_tx2_slot1, Nfft, Ncp);
        x_tx1_slot2 = ofdm_mod(X_tx1_slot2, Nfft, Ncp);
        x_tx2_slot2 = ofdm_mod(X_tx2_slot2, Nfft, Ncp);

        % 5. Generate 2x2 multipath Rayleigh channel
        h = zeros(Nr, Nt, Lch);
        for rx = 1:Nr
            for tx = 1:Nt
                h(rx, tx, :) = sqrt(pdp/2) .* (randn(1, Lch) + 1j*randn(1, Lch));
            end
```

```
    end

    % 6. Channel transmission + AWGN
    y_slot1 = zeros(Nr, Nfft + Ncp + Lch - 1);
    y_slot2 = zeros(Nr, Nfft + Ncp + Lch - 1);

    for rx = 1:Nr
        h_rx_tx1 = squeeze(h(rx, 1, :)).';
        h_rx_tx2 = squeeze(h(rx, 2, :)).';

        y_slot1(rx, :) = conv(x_tx1_slot1, h_rx_tx1) + conv(x_tx2_slot1, h_rx_tx2);
        y_slot2(rx, :) = conv(x_tx1_slot2, h_rx_tx1) + conv(x_tx2_slot2, h_rx_tx2);
    end

    % Add AWGN
    noise1 = sqrt(noiseVar/2) * (randn(size(y_slot1)) + 1j*randn(size(y_slot1)));
    noise2 = sqrt(noiseVar/2) * (randn(size(y_slot2)) + 1j*randn(size(y_slot2)));
    y_slot1 = y_slot1 + noise1;
    y_slot2 = y_slot2 + noise2;

    % 7. OFDM receiver: remove CP + FFT
    Y1 = zeros(Nr, Nfft);
    Y2 = zeros(Nr, Nfft);
    for rx = 1:Nr
        Y1(rx, :) = ofdm_demod(y_slot1(rx, :), Nfft, Ncp);
        Y2(rx, :) = ofdm_demod(y_slot2(rx, :), Nfft, Ncp);
    end

    % 8. Channel frequency response
    H = zeros(Nr, Nt, Nfft);
    for rx = 1:Nr
        for tx = 1:Nt
            h_temp = squeeze(h(rx, tx, :)).';
            H(rx, tx, :) = fft([h_temp zeros(1, Nfft - Lch)], Nfft);
        end
    end
    G = txScale * H;             % Effective channel response; absorbs transmit scale factor
```

```

    % 9. Alamouti STBC decoding (2 receive antennas)
    s1_hat = zeros(1, Nfft);
    s2_hat = zeros(1, Nfft);

    for sc = 1:Nfft
        numerator_s1 = 0;
        numerator_s2 = 0;
        denominator = 0;

        for rx = 1:Nr
            g1 = G(rx, 1, sc);
            g2 = G(rx, 2, sc);
            y1 = Y1(rx, sc);
            y2 = Y2(rx, sc);

            numerator_s1 = numerator_s1 + conj(g1)*y1 + g2*conj(y2
);
            numerator_s2 = numerator_s2 + conj(g2)*y1 - g1*conj(y2
);
            denominator = denominator + abs(g1)^2 + abs(g2)^2;
        end

        % Note: txScale^2 cancels out in numerator/denominator,
restoring constellation scale
        s1_hat(sc) = numerator_s1 / denominator;
        s2_hat(sc) = numerator_s2 / denominator;
    end

    rxSymbols = [s1_hat.'; s2_hat.'];

    % 10. 16-QAM demodulation
    rxBits = qam16_demod(rxSymbols);

    % 11. Error metrics calculation
    bitErrors = sum(txBits ~= rxBits);
    txBitMatrix = reshape(txBits, k, []).';
    rxBitMatrix = reshape(rxBits, k, []).';
    symErrors = sum(any(txBitMatrix ~= rxBitMatrix, 2));

    totalBitErrors = totalBitErrors + bitErrors;
    totalSymErrors = totalSymErrors + symErrors;
    totalBits = totalBits + NbitsPerBlock;
```

```
        totalSymbols = totalSymbols + NsymPerBlock;
    end

    BER(isnr) = totalBitErrors / totalBits;
    SER(isnr) = totalSymErrors / totalSymbols;

    fprintf('Eb/N0 = %2d dB, BER = %.5e, SER = %.5e\n', ...
        EbN0_dB(isnr), BER(isnr), SER(isnr));
end

%% ============================================================
% 12. Plot BER and SER Results
% ============================================================
figure;
semilogy(EbN0_dB, BER, 'o-', 'LineWidth', 1.5, 'MarkerFaceColor', 'b')
    ;
hold on;
semilogy(EbN0_dB, SER, 's-', 'LineWidth', 1.5, 'MarkerFaceColor', 'r')
    ;
grid on;
xlabel('E_b/N_0 (dB)');
ylabel('Error Probability');
title('Performance of 2x2 MIMO-OFDM with Alamouti STBC and 16-QAM');
legend('BER', 'SER', 'Location', 'southwest');

%% ============================================================
% Local functions
% ============================================================

function symbols = qam16_mod(bits)
    bits = bits(:);
    if mod(length(bits), 4) ~= 0
        error('Number of bits must be a multiple of 4 for 16-QAM.');
    end
    bitGroups = reshape(bits, 4, []).';
    bI = bitGroups(:, 1:2);
    bQ = bitGroups(:, 3:4);

    I = bits_to_gray_level(bI);
    Q = bits_to_gray_level(bQ);
    symbols = (I + 1j*Q) / sqrt(10);    % Es = 1 normalization
end

```

```
function bits = qam16_demod(symbols)
    symbols = symbols(:) * sqrt(10);
    I = real(symbols);
    Q = imag(symbols);

    bitsI = gray_level_to_bits(I);
    bitsQ = gray_level_to_bits(Q);
    bitGroups = [bitsI bitsQ];
    bits = reshape(bitGroups.', [], 1);
end

function levels = bits_to_gray_level(bitPairs)
    % Gray mapping: 00 -> -3, 01 -> -1, 11 -> +1, 10 -> +3
    levels = zeros(size(bitPairs, 1), 1);
    for i = 1:size(bitPairs, 1)
        b1 = bitPairs(i, 1);
        b2 = bitPairs(i, 2);
        if b1 == 0 && b2 == 0,     levels(i) = -3;
        elseif b1 == 0 && b2 == 1, levels(i) = -1;
        elseif b1 == 1 && b2 == 1, levels(i) = 1;
        elseif b1 == 1 && b2 == 0, levels(i) = 3;
        end
    end
end

function bitPairs = gray_level_to_bits(values)
    % Gray demapping decision boundaries
    bitPairs = zeros(length(values), 2);
    for i = 1:length(values)
        v = values(i);
        if v < -2,     bitPairs(i, :) = [0 0];
        elseif v < 0,  bitPairs(i, :) = [0 1];
        elseif v < 2,  bitPairs(i, :) = [1 1];
        else,          bitPairs(i, :) = [1 0];
        end
    end
end

function txTime = ofdm_mod(X, Nfft, Ncp)
    x = ifft(X, Nfft) * sqrt(Nfft);
    txTime = [x(end-Ncp+1:end) x];
end
```

```
function X = ofdm_demod(rxTime, Nfft, Ncp)
    X = fft(rxTime(Ncp+1:Ncp+Nfft), Nfft) / sqrt(Nfft);
end
```